Task 2: Exploratory Data Analysis

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Explore descriptive statistics, distributions, relationships, category/region performance, and monthly trends. This is synthetic practice data. Unfiltered totals represent recorded sales across completed, returned, and cancelled orders, not net accounting revenue. No causal effects are claimed. Unknown regions remain a data-quality bucket and are excluded from geographic performance rankings.

Code cell 1

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from IPython.display import display
sns.set_theme(style='whitegrid')
# Resolve project root in a repository checkout or a Colab Drive folder.
from pathlib import Path
PROJECT_ROOT = next((p for p in [Path.cwd(), Path.cwd().parent,
    Path('/content/drive/MyDrive/retail-sales-week2')]
    if (p / 'data/raw/retail_sales_raw.csv').is_file()), None)
if PROJECT_ROOT is None:
    raise FileNotFoundError('Open from the extracted retail-sales-week2 folder; see README for Colab setup.')

DATA_PATH = PROJECT_ROOT / 'data/processed/retail_sales_cleaned.csv'
OUTPUT_DIR = PROJECT_ROOT / 'reports/eda'
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
df = pd.read_csv(DATA_PATH, parse_dates=['order_date','ship_date'])
assert not df.isna().any().any()
assert not df.duplicated().any()
print('Dataset:', df.shape)
print('Date range:',df.order_date.min(), 'to',df.order_date.max())
display(df.head())
df.info()
```

```
Dataset: (11923, 21)
Date range: 2023-01-01 00:00:00 to 2025-12-31 00:00:00
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11923 entries, 0 to 11922
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   order_id          11923 non-null  object
 1   order_date        11923 non-null  datetime64[ns]
 2   ship_date         11923 non-null  datetime64[ns]
 3   store_id          11923 non-null  object
 4   region            11923 non-null  object
 5   customer_segment  11923 non-null  object
 6   product_id        11923 non-null  object
 7   product_name      11923 non-null  object
 8   product_category  11923 non-null  object
 9   quantity          11923 non-null  float64
 10  unit_price        11923 non-null  float64
 11  discount          11923 non-null  float64
 12  promotion         11923 non-null  object
 13  sales             11923 non-null  float64
 14  cost_price        11923 non-null  float64
 15  shipping_cost     11923 non-null  float64
 16  profit            11923 non-null  float64
 17  payment_method    11923 non-null  object
 18  sales_channel     11923 non-null  object
 19  holiday_flag      11923 non-null  int64
 20  order_status      11923 non-null  object
dtypes: datetime64[ns](2), float64(7), int64(1), object(11)
memory usage: 1.9+ MB
```

```
    order_id order_date  ship_date store_id   region customer_segment product_id   product_name
product_category  quantity  unit_price  discount      promotion     sales  cost_price  shipping_cost   profit
payment_method sales_channel  holiday_flag order_status
0  ORD000001 2023-04-08 2023-04-10    ST101    north         consumer      P3001  Running Shoes
                 fashion       7.0  3022.96000     0.082  festival sale  19890.12     1610.25          5.960
8612.41            upi      in-store             0     returned
1  ORD000002 2025-04-28 2025-05-01    ST303     east         consumer      P5001   Skincare Kit  beauty &
personal care       3.0  1351.45000     0.107   no promotion   4574.92      657.02        100.970  2502.89
           cash        online             0    completed
2  ORD000003 2024-12-18 2024-12-18    ST401     west         consumer      P2003   Bedsheet Set          home
& kitchen       5.0  1614.89000     0.072   no promotion   9323.44      756.38         27.900  5513.64
credit card      in-store             1    completed
3  ORD000004 2024-04-26 2024-04-26    ST501  central         consumer      P1002     Smartphone
electronics       3.0  6542.12375     0.192   no promotion  47721.03    13601.43         29.195  6874.41
           cash        online             0    completed
4  ORD000005 2024-04-19 2024-04-21    ST301     east         consumer      P1002     Smartphone
```

```
electronics       3.0  1568.26500     0.081   no promotion  50439.35    14503.70          0.000  6928.25
          cash      in-store             0    completed
```

Descriptive statistics

Mean, median, spread, quartiles, and skewness describe numeric columns; frequencies summarize categories.

Code cell 2

```
numeric_cols = df.select_dtypes('number').columns
numeric_stats = df[numeric_cols].describe().T
numeric_stats['median'] = df[numeric_cols].median()
numeric_stats['skewness'] = df[numeric_cols].skew()
display(numeric_stats.round(2))
display(df.describe(include=['object']).T)
numeric_stats.to_csv(OUTPUT_DIR/'descriptive_statistics.csv')
```

```
                count      mean       std       min      25%      50%      75%        max   median  skewness
quantity      11923.0      3.20      1.46      1.00     2.00     3.00     4.00       7.00     3.00      0.53
unit_price    11923.0   2028.28   1807.22    166.32   507.92  1568.26  2920.94    6542.12  1568.26      1.30
discount      11923.0      0.10      0.06      0.00     0.06     0.10     0.14       0.95     0.10      1.07
sales         11923.0  17167.91  46739.36    149.11  1652.59  4334.66  9818.33  519928.33  4334.66      5.09
cost_price    11923.0   3526.40   9293.04    103.67   236.53   785.68  1610.34   47225.34   785.68      3.60
shipping_cost 11923.0     34.94     28.27      0.00    10.35    29.20    58.22     123.23    29.20      0.50
profit        11923.0   6034.42  15451.33 -23369.78   785.53  2108.66  4719.72  265251.76  2108.66      6.61
holiday_flag  11923.0      0.26      0.44      0.00     0.00     0.00     1.00       1.00     0.00      1.09

                count unique           top   freq
order_id        11923  11923     ORD000001      1
store_id        11923     15         ST302    836
region          11923      6         south   2397
customer_segment 11923      4      consumer   8367
product_id      11923     20         P3002    685
product_name    11923     20   Denim Jeans    685
product_category 11923      6       fashion   2631
promotion       11923      4  no promotion   6266
payment_method  11923      6           upi   4450
sales_channel   11923      3      in-store   7366
order_status    11923      3     completed  11226
```

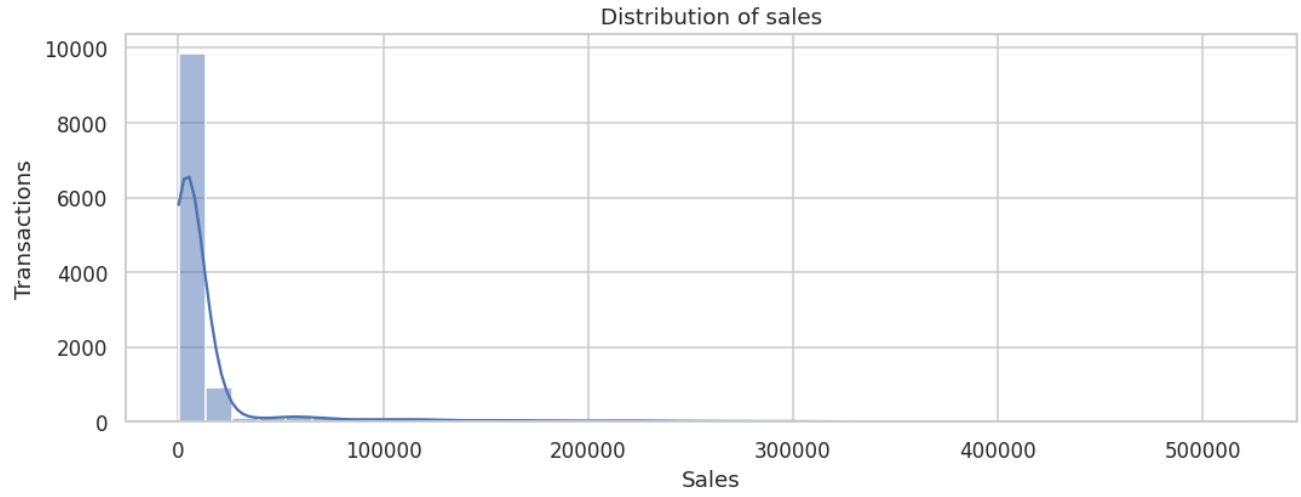
Distributions: histogram and KDE

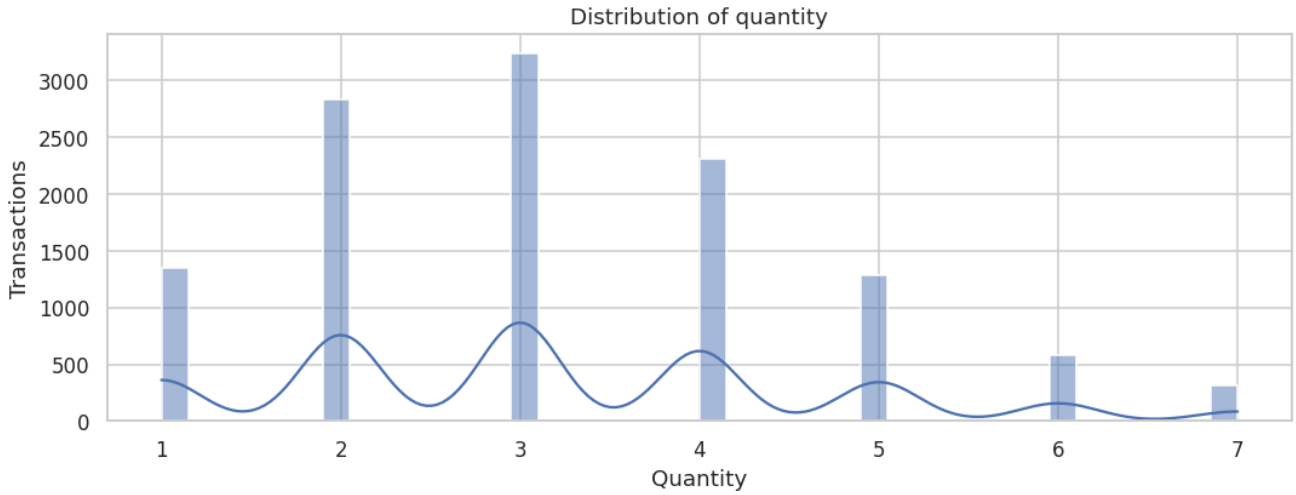
Compare means and medians to assess skew. Extreme values are retained observations, not automatically errors.

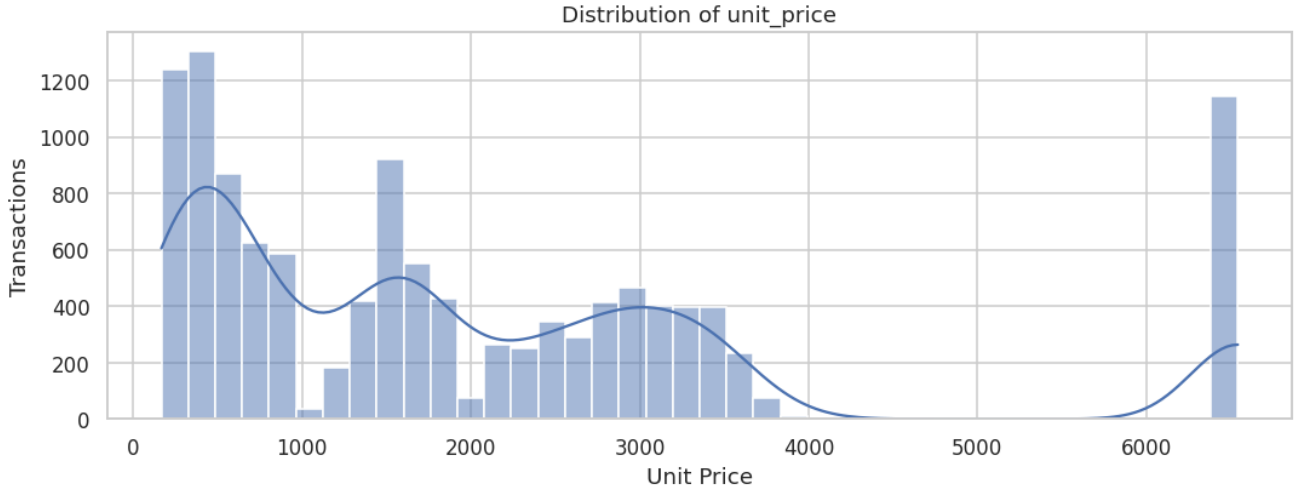
Code cell 3

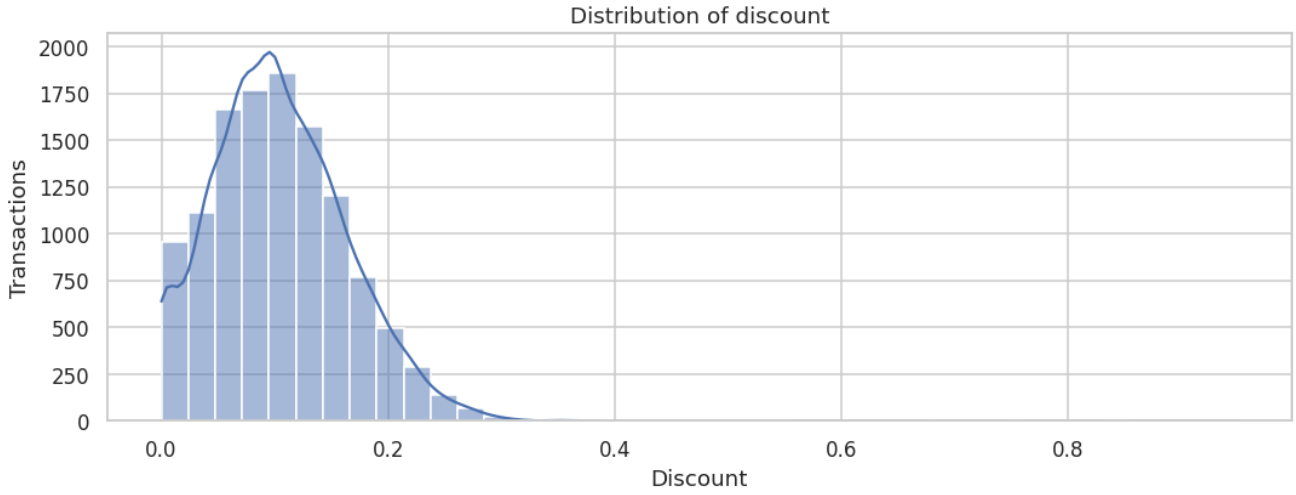
```
for col in ['sales','quantity','unit_price','discount','profit']:
    fig,ax=plt.subplots(figsize=(10,4))
    sns.histplot(data=df,x=col,kde=True,bins=40,ax=ax)
    ax.set(title=f'Distribution of {col}',ylabel='Transactions',xlabel=col.replace('_',' ').title())
    plt.tight_layout();plt.show()
print('Sales mean:',round(df.sales.mean(),2),'| median:',round(df.sales.median(),2),'|
skew:',round(df.sales.skew(),2))
```

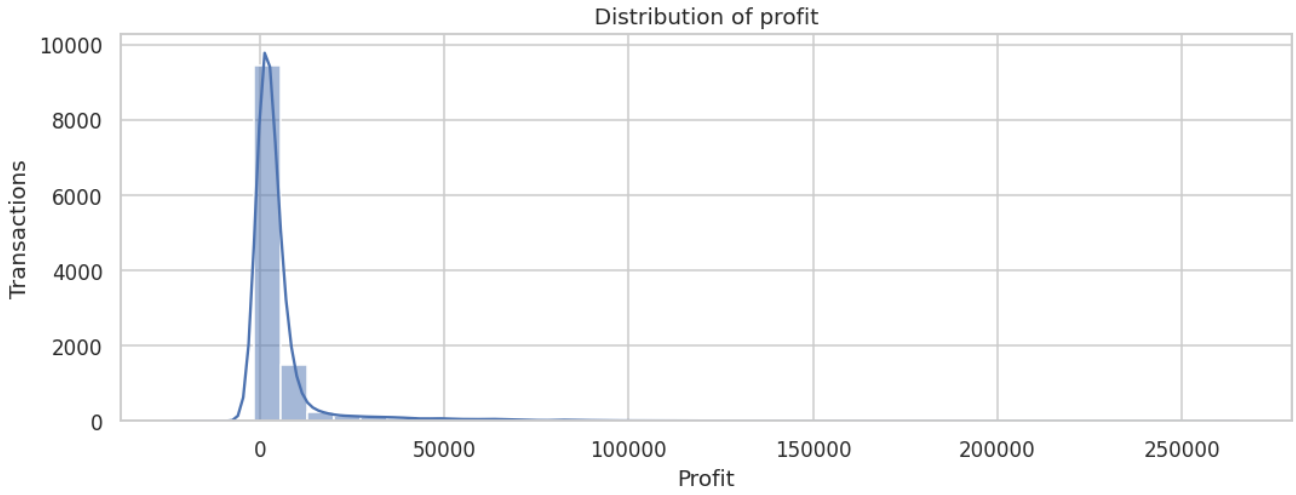
```
Sales mean: 17167.91 | median: 4334.66 | skew: 5.09
```




Distribution of quantity
Transactions
Quantity
0
500
1000
1500
2000
2500
3000
1
2
3
4
5
6
7


Distribution of unit_price
Transactions
Unit Price
0
200
400
600
800
1000
1200
0
1000
2000
3000
4000
5000
6000


Distribution of discount
Transactions
Discount
0
250
500
750
1000
1250
1500
1750
2000
0.0
0.2
0.4
0.6
0.8



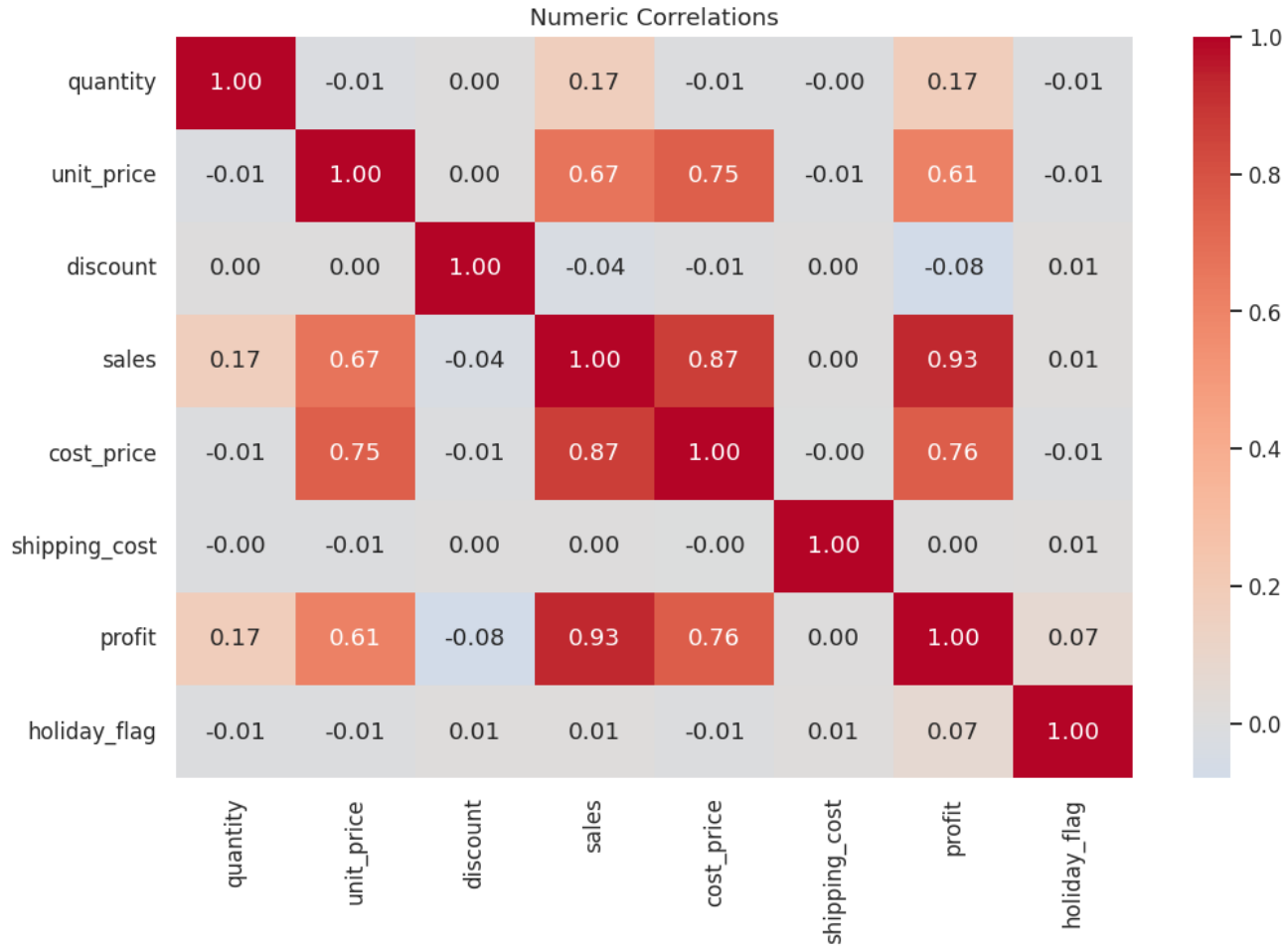
Correlation heatmap

Pearson correlation measures linear association. Profit is derived partly from sales, so their correlation is not independent evidence of causation and profit must not predict sales.

Code cell 4

```
corr_matrix = df[numeric_cols].corr()
plt.figure(figsize=(10,7));sns.heatmap(corr_matrix,annot=True,fmt='.2f',cmap='coolwarm',center=0)
plt.title('Numeric Correlations');plt.tight_layout();plt.show()
corr_matrix.to_csv(OUTPUT_DIR/'correlation_matrix.csv')
print('Discount vs sales correlation:',round(df.discount.corr(df.sales),3))
```

```
Discount vs sales correlation: -0.037
```



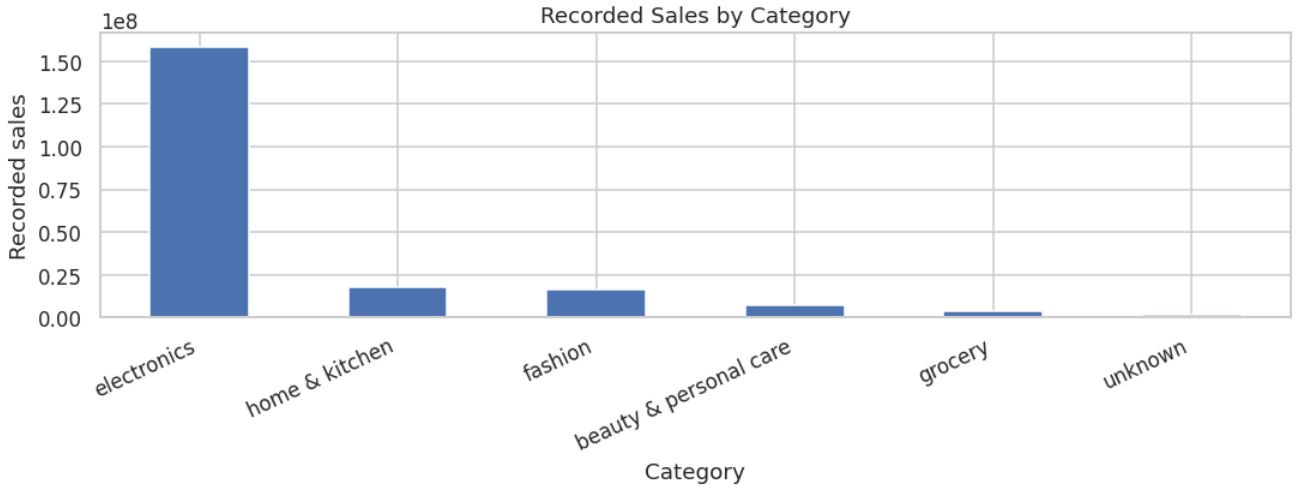
Category performance: bar chart and box plot

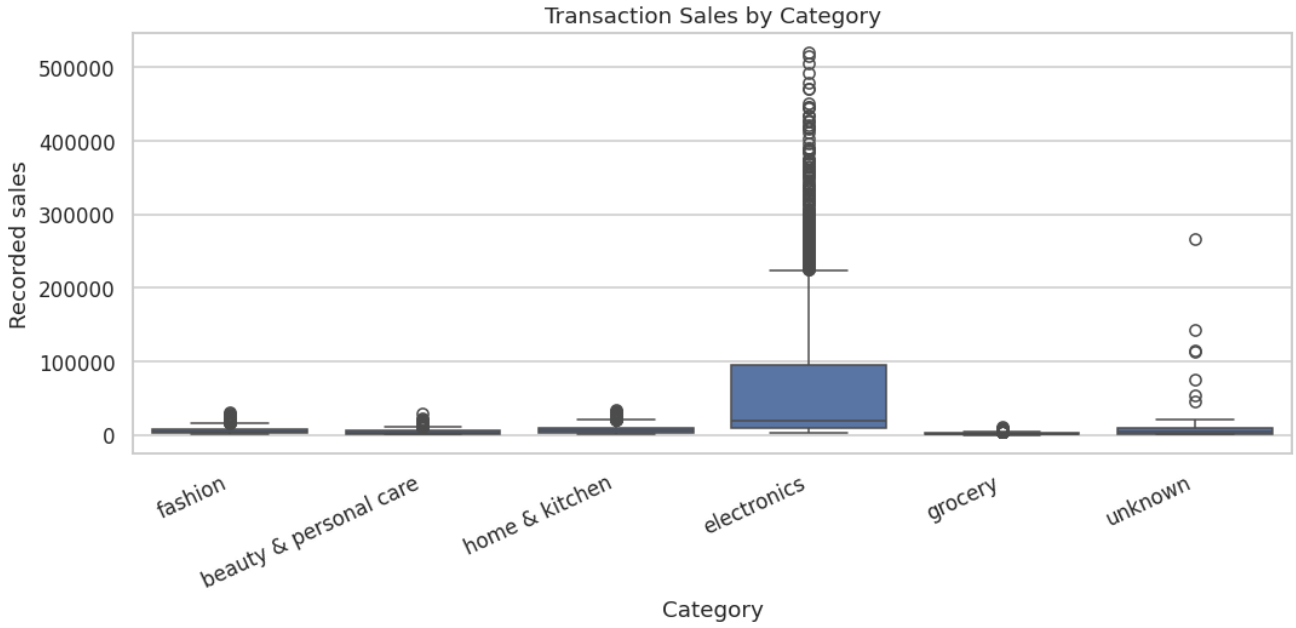
Total sales reflect both order volume and transaction size. Compare totals with transaction counts and distribution.

Code cell 5

```
category_summary = df.groupby('product_category').agg(total_sales=('sales','sum'),average_sales=('sales','mean
'),transactions=('sales','size'),total_profit=('profit','sum')).sort_values('total_sales',ascending=False)
category_summary['sales_share_percent']=100*category_summary.total_sales/df.sales.sum()
display(category_summary.round(2))
category_summary.total_sales.plot.bar(figsize=(10,4),title='Recorded Sales by Category')
plt.ylabel('Recorded
sales');plt.xlabel('Category');plt.xticks(rotation=25,ha='right');plt.tight_layout();plt.show()
plt.figure(figsize=(10,5));sns.boxplot(data=df,x='product_category',y='sales')
plt.title('Transaction Sales by Category');plt.xlabel('Category');plt.ylabel('Recorded
sales');plt.xticks(rotation=25,ha='right');plt.tight_layout();plt.show()
category_summary.to_csv(OUTPUT_DIR/'category_sales_summary.csv')
```

product_category	total_sales	average_sales	transactions	total_profit	sales_share_percent
electronics	1.586342e+08	66934.28	2370	49412880.54	77.50
home & kitchen	1.746812e+07	6987.25	2500	8043625.06	8.53
fashion	1.631883e+07	6202.52	2631	8866790.17	7.97
beauty & personal care	7.075731e+06	3808.25	1858	3683916.81	3.46
grocery	3.746542e+06	1532.33	2445	1407221.32	1.83
unknown	1.449536e+06	12180.98	119	533917.85	0.71





Regional performance

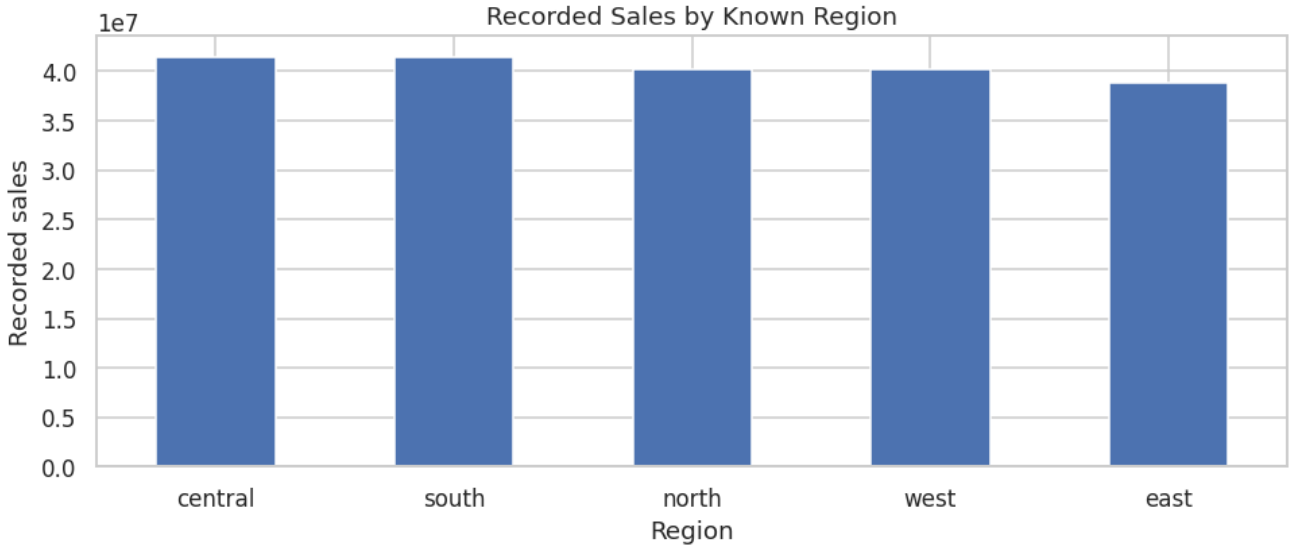
Unknown labels are shown separately and are not a geographic region.

Code cell 6

```
region_summary=df.groupby('region').agg(total_sales=('sales','sum'),transactions=('sales','size'),total_profit
=('profit','sum')).sort_values('total_sales',ascending=False)
known_regions=region_summary.drop(index='unknown',errors='ignore')
display(region_summary.round(2))
known_regions.total_sales.plot.bar(figsize=(9,4),title='Recorded Sales by Known Region')
plt.ylabel('Recorded sales');plt.xlabel('Region');plt.xticks(rotation=0);plt.tight_layout();plt.show()
print('Unknown-region rows:',int(df.region.eq('unknown').sum()))
region_summary.to_csv(OUTPUT_DIR/'regional_sales_summary.csv')
```

```
Unknown-region rows: 144
```

region	total_sales	transactions	total_profit
central	41479793.71	2363	14917043.94
south	41457341.02	2397	14697390.37
north	40243593.89	2322	13622036.90
west	40212732.37	2339	14540886.10
east	38866345.03	2358	13326212.81
unknown	2433185.13	144	844781.63

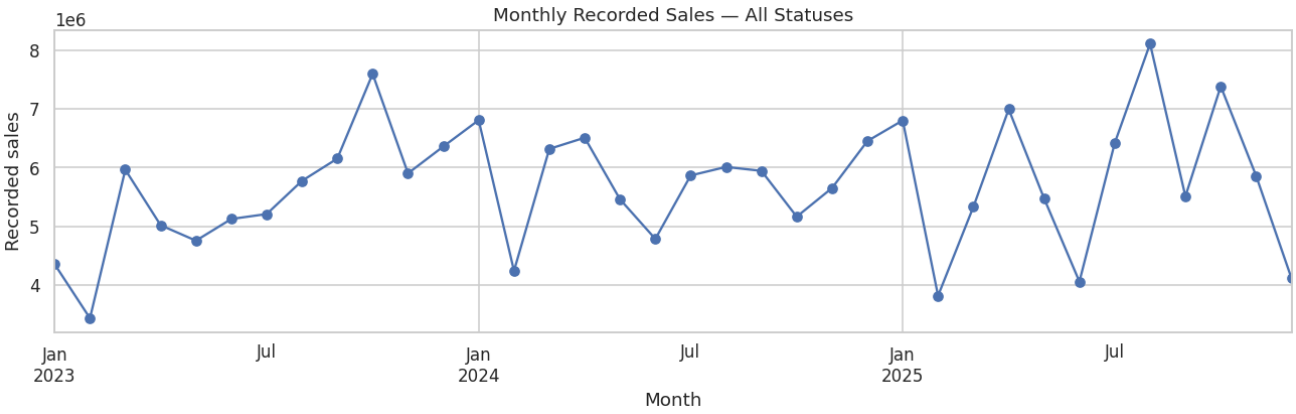


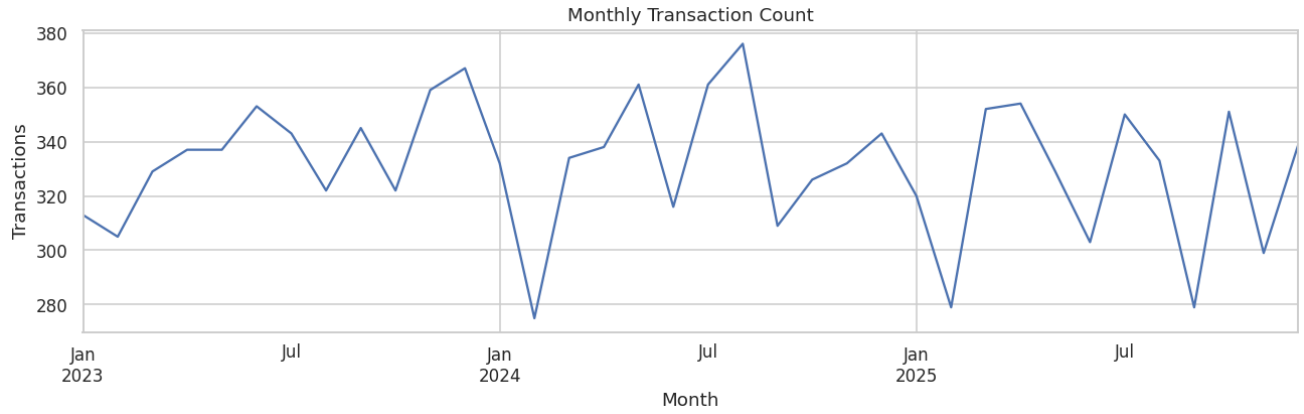
Monthly trends: line chart

Peaks may reflect order volume or product mix; this chart alone does not establish stable seasonality.

Code cell 7

```
monthly_summary=df.set_index('order_date').resample('MS').agg(total_sales=('sales','sum'),transactions=('sales
','size'),average_sales=('sales','mean'))
monthly_summary.total_sales.plot(figsize=(12,4),title='Monthly Recorded Sales — All Statuses',marker='o')
plt.ylabel('Recorded sales');plt.xlabel('Month');plt.tight_layout();plt.show()
monthly_summary.transactions.plot(figsize=(12,4),title='Monthly Transaction Count')
plt.ylabel('Transactions');plt.xlabel('Month');plt.tight_layout();plt.show()
monthly_summary.to_csv(OUTPUT_DIR/'monthly_sales_summary.csv')
```



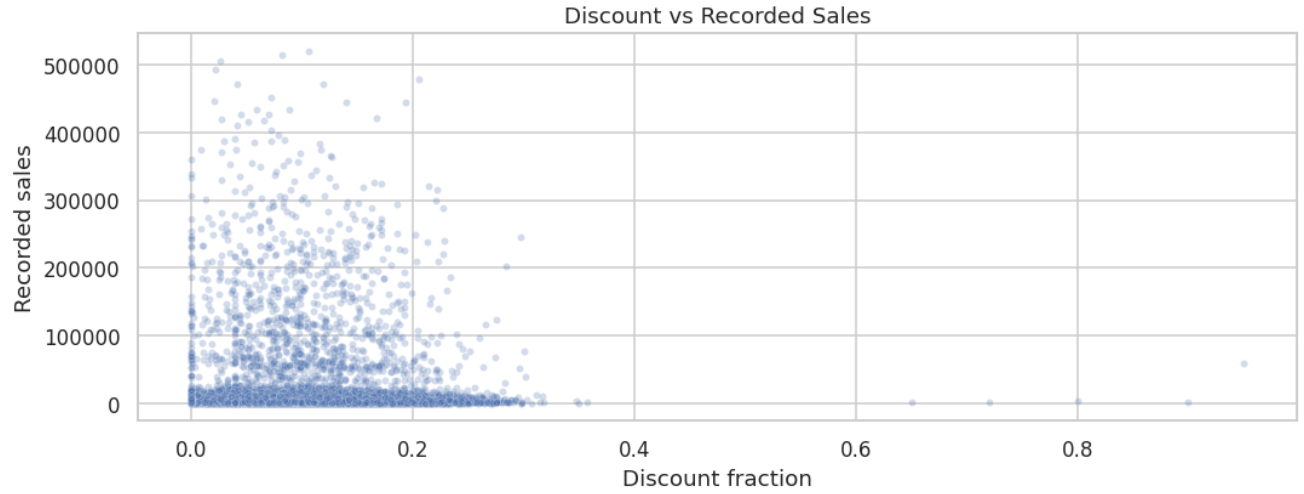


Discount and sales: scatter plot

Assess the point cloud and correlation; product mix confounds comparisons.

Code cell 8

```
plt.figure(figsize=(10,4));sns.scatterplot(data=df,x='discount',y='sales',alpha=.25,s=15)
plt.title('Discount vs Recorded Sales');plt.xlabel('Discount fraction');plt.ylabel('Recorded
sales');plt.tight_layout();plt.show()
discount_df=df.assign(discount_band=pd.cut(df.discount,[-.001,0,.1,.2,.3,1],labels=['No discount','Up to
10%','10–20%','20–30%','Above 30%']))
discount_summary=discount_df.groupby('discount_band',observed=False).agg(total_profit=('profit','sum'),average
_profit=('profit','mean'),transactions=('sales','size'))
display(discount_summary.round(2))
discount_summary.to_csv(OUTPUT_DIR/'discount_band_summary.csv')
```



discount_band	total_profit	average_profit	transactions
No discount	3300323.74	8685.06	380
Up to 10%	39658790.28	6911.61	5738
10–20%	26385120.81	5213.42	5061
20–30%	2542259.67	3492.11	728
Above 30%	61857.25	3866.08	16

Order statuses and data-quality limitations

Completed-only sales are a separate measure. Returns require a defined accounting rule before computing net revenue. Independent cleaning of quantity and price also means transaction arithmetic need not reconcile.

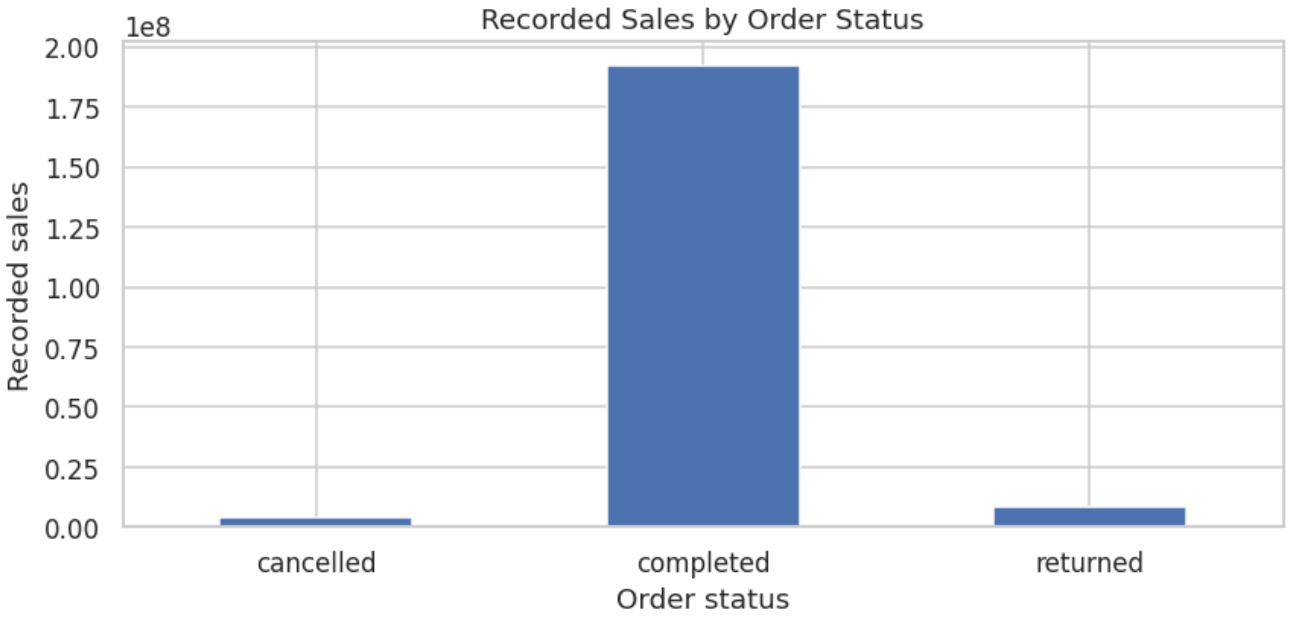
Code cell 9

```
status_summary=df.groupby('order_status').agg(transactions=('sales','size'),recorded_sales=('sales','sum'),rec
orded_profit=('profit','sum'))
display(status_summary.round(2))
status_summary.recorded_sales.plot.bar(figsize=(8,4),title='Recorded Sales by Order Status')
plt.ylabel('Recorded sales');plt.xlabel('Order status');plt.xticks(rotation=0);plt.tight_layout();plt.show()
completed=df[df.order_status.eq('completed')]
```

```
print('Completed-only sales:',f'{completed.sales.sum():,.2f}')
print('Exact duplicate rows:',df.duplicated().sum(),'| repeated IDs:',df.order_id.duplicated().sum())
print('Records not reconciling with quantity × price × (1-discount), tolerance 1:',int((abs(df.sales-
df.quantity*df.unit_price*(1-df.discount))>1).sum()))
status_summary.to_csv(OUTPUT_DIR/'status_sales_summary.csv')
```

```
Completed-only sales: 192,427,623.40
Exact duplicate rows: 0 | repeated IDs: 0
Records not reconciling with quantity × price × (1-discount), tolerance 1: 11897
```

order_status	transactions	recorded_sales	recorded_profit
cancelled	217	3.980776e+06	1457672.20
completed	11226	1.924276e+08	67400102.57
returned	480	8.284592e+06	3090576.98



Findings and recommendations

The following insights are computed from the current cleaned CSV, so they remain consistent after rerunning Task 1.

Code cell 10

```
top=category_summary.iloc[0]
low=known_regions.iloc[-1]
peak=monthly_summary.total_sales.idxmax()
band=discount_summary.loc[discount_summary.transactions.gt(0),'average_profit'].idxmax()
insights=[
 f'1. {category_summary.index[0]} contributes {top.sales_share_percent:.2f}% of recorded sales. Review product
 mix, unusually large transactions, and profitability before prioritizing stock.',
 f'2. {known_regions.index[-1]} has the lowest known-region recorded sales ({low.total_sales:,.2f}). Compare
transaction counts and product mix before changing budgets; unknown labels are excluded.',
 f'3. {band} has the highest mean recorded profit per transaction. Compare sample size, product mix, and
completed-only outcomes before changing discount policy.',
 f'4. {peak:%B %Y} has the highest monthly recorded sales ({monthly_summary.loc[peak,"total_sales"]:,.2f}).
Investigate promotions, transaction volume, category mix, and order statuses.'
]
for insight in insights:print(insight+'\n')
(OUTPUT_DIR/'insights.md').write_text('# EDA Findings\n\n'+'\n\n'.join(insights)+'\n\nSynthetic observational
data; results are descriptive and do not establish causation. Recorded sales across all statuses are not
realized net revenue.')
```

```
1. electronics contributes 77.50% of recorded sales. Review product mix, unusually large transactions, and
profitability before prioritizing stock.

2. east has the lowest known-region recorded sales (38,866,345.03). Compare transaction counts and product mix
 before changing budgets; unknown labels are excluded.

3. No discount has the highest mean recorded profit per transaction. Compare sample size, product mix, and
completed-only outcomes before changing discount policy.

4. August 2025 has the highest monthly recorded sales (8,106,887.18). Investigate promotions, transaction
volume, category mix, and order statuses.
```

Conclusion

The analysis includes histograms/KDE, a heatmap, bar charts, box plots, line charts, and a scatter plot. Use the computed insights as investigation priorities. Confirm synthetic-data assumptions and transaction formulas, investigate unknown regions, and distinguish recorded totals from completed-order sales. These results describe this dataset; they are not validated business forecasts.